

MATH 415/515 Advanced Linear Algebra

Exam 3 Review

1 List of Topics

1. Exam 1 material

(a) Review

- i. Vector space, V
- ii. Subspace, W
- iii. Linear combination, span, spanning set
- iv. Linear independence
- v. Basis
- vi. Dimension
- vii. Eigenvalue, eigenvector, eigenspace, algebraic multiplicity, geometric multiplicity
- viii. Null space, $\text{NS}(A)$, column space, $\text{CS}(A)$, row space, $\text{RS}(A)$

(b) Inner product on $\mathbb{R}^n$, dot product

- i. Length, norm, $\|\mathbf{v}\|$
- ii. Unit vector, $\hat{\mathbf{v}}$
- iii. Distance, $\|\mathbf{v} - \mathbf{u}\|$
- iv. Angle, θ
- v. Orthogonality
 - A. Orthogonal complement, $W^\perp$
 - B. Orthogonal set, orthogonal basis
 - C. Orthogonal projection of $\mathbf{x}$ onto a subspace W , $\mathbf{x}' = \text{proj}_W(\mathbf{x})$
 - D. The component of $\mathbf{x}$ orthogonal to W , $\mathbf{z}$
 - E. Orthonormal set, orthonormal basis
 - F. Orthogonal matrix, U
 - G. The Gram-Schmidt process
 - H. QR-factorization
 - I. Least-squares solution to $A\mathbf{x} = \mathbf{b}$, normal equations

2. Exam 2 material

(a) Curve fitting

- i. Design matrix, X , parameter vector, β , observation vector, $\mathbf{y}$
- ii. Residual vector, ϵ , the linear regression model
- iii. Coefficient of determination, R^2

(b) Inner product spaces

(c) Fourier series

(d) Orthogonal polynomials

- i. Inner product, orthogonality

(e) Applications to linear differential equations

- i. Linear transformation, kernel, eigenvalue, eigenfunction
- ii. Sturm-Liouville problems
- iii. Bessel functions

- iv. Orthogonal polynomials: Legendre, Chebyshev, Hermite, Laguerre, Jacobi
- (f) [MATH 515] QR algorithm
- 3. Symmetric matrices and orthogonal diagonalizability
- 4. Singular value decomposition
- 5. [MATH 515] Jordan canonical form

2 Sample Questions

Short-answer questions for Exam 2 may come from the following list. If I think of additional problems, I reserve the right to include them on Exam 2.

Review (Exam 1)

1. Define a subspace W of a vector space V .
2. Define a linear combination of the vectors $\{\mathbf{v}_1, \dots, \mathbf{v}_p\}$.
3. Define $\text{Span}\{\mathbf{v}_1, \dots, \mathbf{v}_p\}$.
4. Define linear independence for $\{\mathbf{v}_1, \dots, \mathbf{v}_p\}$.
5. Define a basis of a subspace W .
6. Define the dimension of a subspace W . Include definitions for both a finite-dimensional subspace and an infinite-dimensional subspace.
7. State the Basis Theorem.
8. Define an eigenvalue, λ , and an associated eigenvector, $\mathbf{v}$, of a matrix A .
9. State a theorem relating the concepts of eigenvectors and linear independence.
10. State the definition of a linear transformation, $T : V \rightarrow W$.

Inner product (Exam 1)

11. Define the inner product on $\mathbb{R}^n$.
12. Define orthogonality between two vectors.

Orthogonal sets (Exam 1)

13. Define an orthogonal basis for a subspace W .
14. Define the orthogonal projection of $\mathbf{x}$ onto a subspace W , $\text{proj}_W(\mathbf{x})$.
15. Define an orthonormal basis for a subspace W .

Orthogonal projections (Exam 1)

16. State the Orthogonal Decomposition Theorem.
17. State the Best Approximation Theorem.

The Gram-Schmidt process (Exam 1)

18. Explain the purpose of the Gram-Schmidt process. (What do we need before we can start the process and what does it produce?)
19. How does the Gram-Schmidt process work? (What happens at each stage of the process? A picture might help.)

Least-squares problems (Exam 1)

20. Define a least-squares solution of $A\mathbf{x} = \mathbf{b}$.
21. Define the normal equations for $A\mathbf{x} = \mathbf{b}$.
22. How do we calculate the error associated with a least-squares solution?

Curve fitting (Exam 2)

23. State the linear regression model, and label all four pieces of the model.
24. Explain why even though a curve might not be linear, the regression model is still linear.
25. Be able to fit a curve to data.
26. Explain what the coefficient of determination, R^2 , measures. What are the range of values, and what do those values mean?
27. What two cases must be considered when calculating the coefficient of determination, R^2 ?

Inner product spaces (Exam 2)

28. Define a general inner product.
29. Define an inner product for $\mathbb{C}^n$.
30. Give two examples of an inner product that are not the usual inner product involving the dot product.
31. Be able to perform any of the eight calculations outlined as applications of an inner product for a given vector space and inner product.

Fourier series and orthogonal polynomials (Exam 2)

32. What is a Fourier series? Why is it a topic in an advanced linear algebra course?
33. Be able to calculate a Fourier series with technology.
34. Define a sequence of orthogonal polynomials.
35. State a theorem that provides information about a sequence of orthogonal polynomials and its use as a basis and connection with the orthogonal complement.

Applications for linear differential equations (Exam 2)

36. How does solving a linear, homogeneous ODE relate to finding the kernel of a linear transformation?
37. How does solving a linear, homogeneous ODE relate to eigenvalues and eigenfunctions?
38. Define a Sturm-Liouville problem. Define a regular Sturm-Liouville problem.
39. Be able to use the μ -integrating factor method to put an ODE into Sturm-Liouville form and identify $w(x)$.
40. State a theorem that provides six facts about regular Sturm-Liouville problems.
41. Explain why Sturm-Liouville theory, which deals with ODEs, is being discussed in an advanced linear algebra course.
42. What are Bessel functions? How are they similar and different from the families of orthogonal polynomials discussed, such as Legendre, Chebyshev, and Hermite polynomials?
43. Be able to take a Sturm-Liouville problem, find its inner product, its eigenvalues, and use Taylor series to find its associated eigenfunctions.
44. In the process for finding various polynomials, we assumed $y = \sum a_n x^n$. Walk through that process and explain every place where linear algebra is used.

Symmetric matrices

45. Define an orthogonal diagonalization of the matrix A .

46. State an if-and-only-if theorem concerning orthogonal diagonalization.
47. Be able to orthogonally diagonalize a matrix.
48. State the Spectral Theorem.
49. Be able to prove the Spectral Theorem, given the existence of a real Schur factorization. ((1), (2), and (4) are Problems 7, 9, and 10 of Homework #9. (3) is in the notes.)

Singular Value Decomposition

50. Define the singular values of A .
51. State the Singular Value Decomposition Theorem.
52. Be able to calculate the singular value decomposition of a matrix.
53. Explain how the final four entries to the Invertible Matrix Theorem can be proven from the singular value decomposition. The diagram from the final day of class will be useful.

Jordan Canonical Form [MATH 515]

54. Define a Jordan block, $J_m(\lambda)$.
55. Define a generalized eigenvector of the eigenvalue of λ .
56. How do we determine the size of the largest Jordan block for a given eigenvalue? How do we determine the number of Jordan blocks for an eigenvalue?
57. Explain how to calculate a chain of generalized eigenvectors for a Jordan block for the eigenvalue of λ .
58. Define the Jordan canonical form for a matrix A .
59. Be able to calculate the Jordan canonical form of a matrix.

Technology note

60. Be prepared to use your laptop and Maple and MATLAB to compute things like Fourier series, Gram-Schmidt with general inner products, finding the best approximation with general inner products, fitting data to curves, etc. You also need to have a plan for submitting those files and calculations to a D2L dropbox.

If you have code from the homework that you would like to use, say for computing least squares, fitting data to a polynomial, or calculating the coefficient of determination, you are free to use that. Just specify that you are doing so on the exam. Also, this means that if your code had any issues from the homework, now would be the time to fix that code.

On the exam, though, any calculations must start with a blank file.

3 Practice Problems

1. (T/F) questions. Justify your answer.
 - (a) There are symmetric matrices that are not orthogonally diagonalizable.
 - (b) An orthogonal matrix is orthogonally diagonalizable.
 - (c) If $A^T = A$ and if vectors $\mathbf{u}$ and $\mathbf{v}$ satisfy $A\mathbf{u} = 3\mathbf{u}$ and $A\mathbf{v} = 4\mathbf{v}$, then $\mathbf{u} \cdot \mathbf{v} = 0$.
2. Prove that if 0 is an eigenvalue of $A^T A$, then the columns of A are linearly dependent. Hint: consider an eigenvector of the eigenvalue 0, $\mathbf{x}$, and compute $\mathbf{x}^T A^T A \mathbf{x}$. The follow the definition of linear dependence.
3. Assume that A is an $n \times 2$ matrix with linearly dependent columns, or $A = [\mathbf{v} \ k\mathbf{v}]$, where $\mathbf{v} \neq \mathbf{0}$ and $k \neq 0$. Prove that the singular values of A are 0 and $\|\mathbf{v}\| \sqrt{1 + k^2}$.
4. Prove that if P is an orthogonal $m \times m$ matrix and A is $m \times n$, then PA and A have the same singular values.
5. Suppose that A is square and invertible. Find a singular value decomposition of A^{-1} .

Theorem (The Invertible Matrix Theorem). Let A be an $n \times n$ matrix. The following are equivalent.

- (1) A is an invertible matrix.
- (2) A is row equivalent to the $n \times n$ identity matrix.
- (3) A has n pivot rows.
- (4) The equation $A\mathbf{x} = \mathbf{0}$ has only the trivial solution.
- (5) The columns of A form a linearly independent set.
- (6) The linear transformation $\mathbf{x} \mapsto A\mathbf{x}$ is one-to-one/injective.
- (7) The equation $A\mathbf{x} = \mathbf{b}$ has at least one solution for each $\mathbf{b} \in \mathbb{R}^n$.
- (8) The columns of A span $\mathbb{R}^n$.
- (9) The linear transformation $x \mapsto A\mathbf{x}$ is onto/surjective.
- (10) There is an $n \times n$ matrix C such that $CA = I$.
- (11) There is an $n \times n$ matrix D such that $AD = I$.
- (12) A^T is an invertible matrix.
- (13) $\det(A) \neq 0$.
- (14) The columns of A form a basis of $\mathbb{R}^n$.
- (15) $\text{CS}(A) = \mathbb{R}^n$.
- (16) $\text{rk}(A) = n$.
- (17) $\nu(A) = 0$.
- (18) $\text{NS}(A) = \{\mathbf{0}\}$.
- (19) The number 0 is not an eigenvalue of A .
- (20) $(\text{CS}(A))^\perp = \{\mathbf{0}\}$.
- (21) $(\text{NS}(A))^\perp = \mathbb{R}^n$.
- (22) $\text{RS}(A) = \mathbb{R}^n$.
- (23) A has n nonzero singular values.